

SeokHyeon (Lucas) Kong

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RESEARCH INTERESTS

Quantum Computer Systems and Architectures, Noisy Intermediate-Scale Quantum, Variational Quantum Algorithms, Fault-Tolerant Quantum Computers, Quantum Error Correction

EDUCATION

Sungkyunkwan University (SKKU)

College of Information and Communication Engineering

Bachelor of Science in Electronic and Electrical Engineering

Bachelor of Science in Advanced Semiconductor Engineering

Seoul, South Korea

Expected February 2026

GPA: 3.95/4.0 (4.37/4.5)

System Architecture Track

The Pennsylvania State University

School of Electrical Engineering and Computer Science (EECS)

Exchange Student in Computer Engineering

University Park, PA

Fall 2024

Grade: 4.0/4.0

PUBLICATIONS

S. H. Kong, D. Kim, K. Maeng*, and E. Seo*, “Characterizing the system overhead of discrete Gaussian noise generation for differential privacy,” *IEEE Computer Architecture Letters*, in preparation, 2025

S. H. Kong and H. Oh*, “Predicting key regional real estate prices using machine learning technique: with an emphasis on Jeonsae system,” *Journal of the Korean Institute of Information and Communication Engineering*, vol. 27, no. 10, pp. 1208–1213, Oct. 2023

RESEARCH EXPERIENCE

Research Mentee

IonQ, Advised by Dr. Denny Dahl

Remote

08/2025-09/2025

- Addressed the Instant Insanity problem on NISQ devices by employing a graph-theoretic encoding and a divide-and-conquer approach with two distinct post-processing methods.
- Applied a Variational Quantum Algorithm to the Four Corners Map Coloring problem.
- Presented two research posters in English on Constraint Satisfaction Problems in NISQ devices at Qcenter (Quantum Information Research Support Center), Korea.

Undergraduate Research Intern

ArchLab @ SKKU, Advised by Prof. Dongmoon Min

Suwon, South Korea

07/2025-08/2025

- Built foundational expertise in Quantum / Cryogenic Computer Systems through technical presentations and academic defenses.
- Explored cryogenic computer architecture - processors, interconnects, caches, and memory - as well as quantum computer systems with a focus on superconducting qubits.

Undergraduate Researcher

Penn State CSE, Advised by Prof. Kiwan Maeng

University Park, PA

08/2024-09/2025

- Conducted in-depth GPU profiling with Nsight Compute and Nsight Systems to analyze performance bottlenecks and system-level overhead in differentially private training.
- Proposed an optimization that resolves the dominant Geometric operation overhead while preserving Gaussian noise quality.
- Evaluated the real-world applicability of discrete Gaussian noise through Differentially Private Stochastic Gradient Descent (DP-SGD) experiments across multiple models.

Undergraduate Research Intern

Pohang, South Korea

POSTECH QCQN (Quantum Computing and Quantum Networks)

01/2025

- Performed 397 nm fluorescence (Doppler) and 729 nm quadrupole spectroscopy on Ca^+ ions and enhanced system stability through targeted adjustments.
- Implemented micromotion optimization via 2D scanning with DC compensation rods to enhance ion trapping stability.

HONORS & AWARDS

Quantum Hackathon - Director's Award 2025

President's List (Honor Society at SKKU, **2 consecutive years**) 2024, 2025

Dean's List (**5 consecutive semesters**) SP2023, FA2023, SP2024, FA2024, SP2025

Korea-U.S. Student Exchange Scholarship in the field of high-tech industry (\$9,000) FA2024

Student Success Scholarship 2023

TEACHING EXPERIENCE

Academic Tutor

Suwon, South Korea

Sungkyunkwan University (SKKU), Intro to Electromagnetism

09/2023-12/2023

- Taught Samsung Semiconductor blue-collar workers studying Material-Component Engineering at SKKU, with a primary focus on problems not covered in class.

International Summer School Teaching Assistant

Seoul, South Korea

Sungkyunkwan University (SKKU), Engineering Mathematics 2

07/2023-08/2023

- Prepared and organized class materials, including exams.

TECHNICAL SKILLS

GPU Profiling (Nsight Systems, Nsight Compute), Circuit (Cadence Virtuoso, SoC, Verilog), Programming (C, Python, CUDA, MATLAB), Quantum Computing (Qiskit), Data Analysis (NumPy, Pandas, Matplotlib), Scientific Writing (LaTeX)

CERTIFICATIONS

Qiskit Global Summer School 2025 Quantum Excellence (IBM Quantum)

CS50's Introduction to Artificial Intelligence with Python (Harvard University)

CS50's Introduction to Cybersecurity (Harvard University)